

Midterm for Electromagnetic Theory (PHYS330)

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Abstract

This exam may be completed at home, and covers chapters 1-4 of the course text and in-class examples. Class notes and the course text may be used for review. The course warm-up exercises are good study materials for this exam.

1 Chapter 1

- (a) If $\mathbf{A}$ and $\mathbf{B}$ are two vector functions, what does the expression $(\mathbf{A} \cdot \nabla)\mathbf{B}$ mean? Derive $(\hat{\mathbf{r}} \cdot \nabla)\hat{\mathbf{r}}$, where $\hat{\mathbf{r}}$ is $\mathbf{r}/r$. (b) One can show that the *force* on a dipole moment $\mathbf{p}$ induced by a non-uniform field is

$$\mathbf{F} = (\mathbf{p} \cdot \nabla)\mathbf{E} \quad (1)$$

Compute the force on a physical dipole located at the origin with $\mathbf{p} = q\mathbf{d}$ in a field with associated potential $V(\mathbf{r}) = V_0 r^2 + V_1$.

- Evaluate the following integral using (a) the three-dimensional Dirac delta function, or (b) integration by parts. Solving both earns a bonus point.

$$J = \int_{\mathcal{V}} e^{-r} \left(\nabla \cdot \frac{\hat{\mathbf{r}}}{r^2} \right) \quad (2)$$

2 Chapter 2

- Suppose two dipoles, each with dipole moment $\mathbf{p}$ pointed in opposite directions, form a square with alternating positive and negative charges and side length d . Calculate the field $\mathbf{E}_{\text{tot}}$ at the following points P : (a) $P = (0, 0)$, (b) $P = (2d, 0)$, and $P = (0, 2d)$. Check units and take limits¹.
- The electric potential of some configuration is given by the expression

$$V(\mathbf{r}) = A \frac{e^{-\lambda r}}{r} \quad (3)$$

In Eq. 3, A and λ are constants. Find the field $\mathbf{E}(\mathbf{r})$, the charge density ρ and the total charge Q in terms of A and λ . *Hint:* $\rho = \epsilon_0 A (4\pi\delta^3(\mathbf{r}) - \lambda^2 \exp(-\lambda r)/r)$. **Bonus:** compute the total energy stored in the field over all space.

- (a) Use Gauss' Law to compute the field $\mathbf{E}$ as a function of the distance s from a long straight wire with positive charge density λ . (b) Calculate the position versus time of a positive point charge q with mass m if it is released a distance s from the wire.

3 Chapter 3

- Suppose the potential $V_0(\theta)$ at the surface of a sphere of radius R is specified, and there is no charge inside or outside the sphere. (a) Show that the charge density on the sphere is given by

$$\sigma(\theta) = \frac{\epsilon_0}{2R} \sum_{l=0}^{\infty} (2l+1)^2 C_l P_l(\cos \theta) \quad (4)$$

$$C_l = \int_0^\pi V_0(\theta) P_l(\cos \theta) \sin \theta d\theta \quad (5)$$

(b) Produce the specific result for $\sigma(\theta)$ with $V_0(\theta) = P_2(\cos \theta)$.

- A rectangular pipe, running parallel to the z -axis (from $-\infty$ to ∞), has three grounded metal sides, at $y = 0$, $y = a$, and $x = 0$. The fourth side, $x = b$, is maintained at a potential $V_0(y)$. (a) Develop a general formula for the potential inside the pipe. (b) Find the potential explicitly, for the case $V_0(y) = V_0$.
- Suppose a sphere has a uniform polarization $\mathbf{P}$. What are the (a) σ_b and (b) ρ_b ? (c) Which Legendre polynomial does the charge distribution resemble? (d) Calculate the dipole moment of the system. (e) Use the multipole expansion to approximate the voltage far from the sphere. (f) If the system is placed in an external field $\mathbf{E} = E_0 \hat{\mathbf{x}}$, what is the torque?

4 Chapter 4

- (a) Derive the electric field inside a capacitor with surface charge density $\pm\sigma$, assuming the separation between the conductors is small compared to the area. Derive the field for the same system, but with dielectric material between the plates. The bulk polarization of the material is $\mathbf{P} = k\mathbf{E}$.

¹This object is an electrostatic quadrupole.